TAMID QUANT — STRATEGY ONE PAGER

Bitcoin Digital Options: Selling the Kalshi Longshot Tails

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The Idea

Kalshi's hourly "Bitcoin above $K at 5pm ET" contracts overprice longshots in the last hours before expiry. Over 815 expiries (Jul 2024 to Nov 2026), contracts quoted at 1.5-2.5c 3-6h out settled YES 2 times in 3,025 (0.07%). We sell overpriced low-tail YES and buy underpriced high-tail YES, and size each day's book so the worst possible BTC settlement loses at most 3% of bankroll. Until 2026 the 4-5c spreads ate the edge and it almost never traded. Over 481 live days (Jul 2025 to Nov 2026, 237 traded) it made +5.3%, with 7 tail days losing 2.5-3.0% each. The mean daily return, +0.011%, is not significantly above zero (95% CI −0.023% to +0.041%).

Mathematical Framework

Core Model: *A contract pays $1[S_T > K]$. For positions q_i with side s_i (+1 buy, −1 sell YES) at fill p_i, the book's payoff if BTC settles in scenario j is*

$\Pi_j = \Sigma_i\, q_i \cdot [s_i \cdot (1[S_j > K_i] - p_i)]$ − fees. Only m+1 scenarios matter (one between each pair of the day's m strikes), so the worst case $\min_j \Pi_j$ is exact, not simulated.

Signal Construction: *Walk-forward tail probability per (horizon, price bin), using prior expiry days only:*

$f = (\Sigma y + N_0 \cdot m) / (n + N_0)$, $N_0 = 50$, where m is the bin's average market price. It starts at the market price (no edge) and learns from outcomes. Edge = f vs. the sizing price (quote ± 2c slippage, clipped to 1c/99c), less fee.

Entry / Exit Logic: *Decide 6h and 3h before expiry, fill one hour later at the Kalshi bid or ask:*

Trade contracts with mid < 10c or > 90c, spread ≤ 6c, edge ≥ 0.5c and at least 100 prior observations in the bin, after a 30-day warm-up. Hold to settlement.

Position Sizing: *Linear program per expiry day:*

maximize $\Sigma\, q_i \cdot edge_i$ subject to $base_j + \Sigma\, q_i \cdot \Pi_{ij} \geq -3\%$ of bankroll in every scenario j, $0 \leq q_i \leq 50$. Positions already held from the 6h round enter as base. Nested strikes share one budget; a two-sided book (rally and crash risk) gets more room. Whole-contract rounding is re-checked, and 1c of cap is reserved per order for fee rounding.

Key Assumptions

- Tail mispricing seen on earlier days persists; the estimator learns only from past expiries
- Hourly-candle quotes are fillable at 1c / 99c for about 30 contracts (no depth data); fee = 0.07·q·P·(1 − P), rounded up to the cent per order
- Each expiry day is one draw, so the effective sample is 237 traded days, not 402 trades
- 2026's tighter spreads (median 1c in the tails, vs 4-5c before) persist

Key Risks

- Thin margin: +0.47c net per contract (+2.3c won, −93c lost). 1.95% of contracts went against us; breakeven is about 2.4%
- 7 tail days at about −3% (Apr to Oct 2026); each erases about 55 typical traded days. Max drawdown −5.5%
- The cap is sized at the quote, but fills come an hour later, so 24 days ended past 3% (worst 3.14%)
- The earlier 50-day test (+2.4%, no losing day) was lucky. With more history the estimator also trades 5-10c and 90-95c bins, and on the same 50 days it made +1.05% with two tail days
- Per-trade sizing earns +24.8% but lost 8.9% in one day

At a Glance

Asset Class:	*Crypto / Prediction Markets*
Strategy Type:	*Longshot Bias / Tail Risk Premium*
Direction:	*Short Low-Tail YES, Long High-Tail YES*
Holding Period:	*3-6 Hours (Hold to Expiry)*
Universe:	*Kalshi KXBTCD Tail Strikes*

Backtest Results (481 live days)

Sizing	Return	Worst day	Days > 3%
Portfolio cap	+5.3%	−3.0%	24
1 contract / signal	−0.0%	−0.3%	0
Per-trade 3% cap	+24.8%	−8.9%	189

$1,000 bankroll; 3% cap = $30 worst case per expiry day. Jul 18 2025 to Nov 11 2026.

Data Requirements

- Kalshi KXBTCD hourly bid, ask and settlement - public API
- 815 expiries (Jul 2024 to Nov 2026), 36,093 quotes at 6h and 3h; the first year only trains the estimator
- Order-book depth (not yet collected)

Key Parameters (starting values)

- Worst-case cap: 3% of bankroll per expiry day
- Tails: mid < 10c or > 90c; 6h and 3h rounds
- Shrinkage $N_0 = 50$; min 100 obs per bin; 30-day warm-up
- Min edge 0.5c; slippage 2c; max 50 contracts per strike per round

Open Questions

- Does limiting trades to the 1-2.5c / 97.5-99c bins, which hit least often, remove the tail days?
- Does the edge hold out of sample (estimate on 2025, test on 2026)?
- Is there depth to sell 30+ contracts at 1c?
- Can resting (maker) orders beat the 1c / 99c fills and fees?
- Does the same bias exist in Kalshi's hourly S&P 500 contracts (KXINXU)?